

Commentary

Material fallacies of education research evidence and public policy advice

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Introduction

Rational debates about education policy choices are only possible when all those involved use language and evidence correctly. Material fallacies in logic are “mistakes in understanding the meaning or use of terms” (Kreeft, 2010, p. 85). Just as in practical logic, public policy discourse becomes problematic when the terms used are ambiguous, unclear or vague. This commentary analyses one recent case of material fallacy in education policy discourse in order to promote discussion of the responsibilities researchers may have whenever research evidence is used mistakenly.

Treasury’s briefing to the incoming Minister of Finance after the 2011 election included a controversial assertion about school funding policy:

Student achievement can be raised by improving the quality of teaching, which the evidence shows is the largest inschool influence on student outcomes. Increasing student/teacher ratios, and consolidation of the school network, can free up funding that could be used to support initiatives to enhance the quality of teaching, such as more systemic use of value-add data and a more professionalised workforce. (Treasury, 2011, p. 21)

In reply to a Parliamentary question about this briefing, the Minister of Education declined to rule out increases in class size. In short, this was because the “independent observation” of Treasury and the research findings of an influential government adviser, Professor John Hattie, were that schooling policy should instead focus on improving the quality of teaching (House of Representatives, 2012).

Government ministers have many portfolio responsibilities and rely largely on officials and advisers to provide accurate policy advice. At first glance, the Minister’s reply appeared to draw logically on such advice: student/teacher ratios do not have a large influence on student outcomes; the quality of teaching does have a large influence on student outcomes; therefore in order to produce a larger effect on student outcomes, school policy and funding should prioritise the quality of teaching over student/teacher ratios.

Closer analysis reveals errors in both argument and evidence. Much of the terminology is ambiguous and inconsistently used by politicians, officials and academic advisers. The propositions are not demonstrably true—indeed, there is evidence to suggest they are false in crucial respects. The conclusion is, at best,

uncertain because it does not take into account confounding evidence that larger classes *do* adversely affect teaching, learning and student achievement, particularly the achievement of already disadvantaged students such as those prioritised by the Ministry of Education (2011) throughout its current school reform agenda.

Precisely because material fallacies are written or spoken by persons, these limitations give cause to reflect more broadly on the practices, motivations and subject positionings of the various policy actors (politicians, officials and academics) within the contemporary discourse of schooling reform in Aotearoa New Zealand. It has been observed, for example, that, in the English context, the equivalent discourse seeks to portray the public sector as “ineffective, unresponsive, sloppy, risk-averse and innovation-resistant” yet at the same time it promotes “celebration of public sector ‘heroes’ of reform and new kinds of public sector ‘excellence’” (Ball, 2007, p. 3). Relatedly, Mintrom (2000) has written persuasively, in the American context, of the way in which “policy entrepreneurs” position themselves politically to champion, shape and benefit from school reform discourses.

The case discussed in this commentary provides a timely opportunity to ask whether and to what extent New Zealand may too exhibit “a confusing interplay of trust/distrust” (Ball, 2007, p. 3) in the education policy process. It also encourages a more systematic, disinterested analysis of the quality of evidence and advice, and how these are used tactically by various policy actors to facilitate a reformist ideology of public schooling. Finally, it requires us to consider whether researchers have an overriding moral-professional responsibility to challenge well-intentioned reformist education ideologies that are nevertheless based on material fallacies, incomplete research evidence, and invalid conclusions.

There are three principal problems with the Minister’s reliance on Treasury (2011) and Hattie (2009a).

The terms are ambiguous and inconsistently used

The Hattie review was a subjective synthesis of the findings of over 800 meta-analyses conducted by other researchers. Each meta-analysis was categorised according to its influence on student achievement and its average effect size included in a rank order of influences and their effects. Hattie reported the “quality of teaching” (rank 56 of 138) as one of the “contributions” to student achievement from the teacher. However, the term “quality of teaching” was very closely defined. Hattie stated that “all the meta-analyses on the relation of the quality of teaching to learning come from student ratings of teachers by college and university students” (2009a, p. 115). On this definition, the meta-analysis evidence on the quality of teaching has no demonstrable relevance to schooling policy. Moreover, taken literally, it would mean that the Minister and officials assert that achievement should be raised by asking school students to rate their teachers, which is incoherent.

This incoherence suggests that statements by ministers and officials may instead use the term “quality of teaching” loosely, as a slogan (Kreeft, 2010, p. 68). Kreeft states that an expression becomes a slogan “when it is used to produce a thoughtless knee-jerk reaction of agreement or disagreement simply on the basis of the familiarity of

the words rather than on the basis of reason" (p. 78). In this writer's experience, the terms "quality teaching" (Alton-Lee, 2003) and "quality of teaching" (Hattie, 2009a), while originally derived from reasoned analyses of the research literature, have increasingly been deployed as slogans in both scholarly and public policy discourses in this country.

The Treasury quotation above similarly demonstrates an imprecise use of the term because quality of teaching is claimed by Treasury to incorporate "value-add data" and "a professionalised teacher workforce", neither of which is discussed in Hattie's synthesis. Indeed, Treasury appears to have used quality of teaching as a proxy concept (Troyna, 1993 as cited in Gillborn, 1997, p. 352) to gain acceptance for terms that serve quite different, more controversial, policy purposes such as teacher performance management. This absence of clarity does not permit any rational policy debate of either student/teacher ratios or the quality of teaching.

Alternatively, it is possible that the Treasury advice was intended to refer to Hattie's conclusion that a collection of "active and quality teaching strategies" (average effect size 0.68) might have a much greater and more direct combined influence on student learning than various educational structures and working conditions (average effect size 0.08) (2009a, p. 244). Class size is listed among Hattie's working conditions (average effect size 0.21). In addition, the presentation slides of a guest lecture given by the author to Treasury staff based on the *Visible Learning* synthesis (Hattie, 2009b, slides 3 & 4), begin with a graphic scale and table which give the impression that the measured positive effect on student achievement of reducing class size is close to zero. Hence, plausibly, the conclusion among Treasury officials (and thus ministers) could have been that the "quality of teaching" is in practice much more effective at raising school students' achievement than reducing "class size", and that the number of students in a class is consequently insignificant for policy and funding purposes.

There are two problems with such a conclusion. First, it is based on a naive assumption that the two policy variables are mutually exclusive and that consequently one may be manipulated by government independently of the other without causing adverse educational consequences. Second, class size is not simply a working condition or a "classroom compositional effect" to use Hattie's terms (2009a, p. 85). Other kinds of mixed methods analyses based on large-scale observations of what actually takes place in real primary and secondary school classrooms, clearly show that class size materially affects the kinds of practical teaching and learning processes that occur in the classroom. In other words, class size is a "condition" of *both* teaching and learning (e.g., Blatchford, 2003; Blatchford, Russell, & Brown, 2009; Blatchford, Bassett, & Brown, 2011).

Limitations of the research evidence are ignored in the policy advice

Treasury's advice fails to take into account the nature of Hattie's research and the limitations in its applicability to schooling sector policy decisions.

The meta-analyses Hattie reviewed cover all of early childhood education, schooling and college (tertiary) level education. (This is stated at the beginning of the 2009

book by the author but not taken into account for the remainder nor, indeed, at all in its sequel which is targeted specifically at schoolteachers [Hattie, 2012].) Consequently, some of the studies included in (i) the synthesis; (ii) calculations of the average effect size of the studies within a topic category; and (iii) the rank order of effect sizes, are not, in fact, studies of *schooling*. These non-school studies have no proven relevance to the schooling sector and schooling policy decisions. Their inclusion also skews the stated average effect size on achievement for a particular topic and, as a consequence, its position in the rank order (see O'Neill, 2012 for a detailed discussion).

Because the rank order is of average effect sizes across all formal education sectors, with students of all ages in learning settings with very diverse characteristics, it cannot reasonably be claimed to be relevant to policy choices that are specific to the state school sector. A further limitation is that the studies are undertaken with a very broad range of both general and specific student populations. Numerous studies in the synthesis are of students with disabilities and, therefore, of their quite specific educational needs. Yet irrespective of these important contextual details, the studies are combined into a single rank order that is presented as if it were generalisable to the learning needs of all students.

There are several obvious caveats on the use of a table of effect sizes as a practical guide to policy choices. For example, as the author cautions: (a) correlations between particular teacher behaviours and student achievement are not to be confused with the causes of that achievement (Hattie, 2009a, p. 3); and (b) "most of the successful effects come from innovations, and these effects may not be the same as effects of teachers in regular classrooms" (p. 6) (emphasis in original). It is therefore inappropriate for ministers and officials to pick schooling policy solutions directly from a rank ordering of one-dimensional statistical effect sizes and assume that student achievement levels in real classrooms will rise as a result.

Other limitations acknowledged by the author are also ignored by Treasury. It is stated at the outset that the book is "not about classroom life and does not speak to the nuances and details of what happens in classrooms.... It is more concerned with main effects than interactions" (Hattie, 2009a, p. viii). However, real classrooms are all about interactions among variables, and their effects. The author implicitly recognises this when he states that "a review of non-meta-analytic studies could lead to a richer and more nuanced statement of the evidence" (p. 255). Similarly, he acknowledges that when different teaching methods or strategies are used together their combined effects may be much greater than their comparatively small effect measured in isolation (p. 245).

The analysis is concerned only with single variables. Thus if the only difference as a result of class-size change is that the number of students is reduced, but the teacher does nothing differently then it is highly probable that there will be no or very limited measurable effects on student achievement. This, in effect, is the author's own conclusion: "this lack of outcome difference is most likely because teachers do not change their current teaching strategies" (Hattie, 2009a, p. 88). Because the synthesis excludes all qualitative and non-meta-analytic studies it is impossible to make inferences on the basis of the meta-analysis evidence about what may or may

not happen to classroom interactions between teachers and students in larger or smaller classes, or when changes are made to class sizes.

Policy conclusions drawn from the research evidence are invalid

Because the use of terms is unclear and inconsistent, because the limitations of Hattie's research are not taken into account by Treasury, and because other necessary data and evidence are omitted, the conclusions drawn are invalid. Not least, Treasury's advice to increase student/teacher ratios contradicts the author's own advice: "the positive sign of the average effect size suggests that increasing class size is poor policy" (Hattie, 2009a, p. 88).

Treasury advises an increase in future student/teacher ratios but does not include consideration of publicly available data on current ratios. The most recent OECD data (2011, Table D2.2) show that New Zealand ratios are 16.3 in primary and lower-secondary schools and 12.8 in upper secondary schooling. New Zealand ranks 19th out of 35 countries that provided data on primary school ratios, 26th out of 31 on reported lower secondary school ratios and equal 16th out of 31 on reported upper secondary school ratios. These data provide no justification for increasing ratios above their present levels in schools. If there is a logical case of any kind to be made for policy change in this area, it is for New Zealand to adopt the mean OECD primary (*reduce* the NZ ratio to 16.0), and lower secondary (*reduce* to 13.5), and upper secondary (*increase* to 13.5) ratios as maxima.

Hattie explicitly states that his synthesis is not about what occurs in classrooms. One must therefore logically look to other empirical sources for useful research evidence and policy guidance on the interaction or mediation effects between class size and teaching and learning processes. This other evidence reinforces the view that the argument put forward by Treasury for increasing student/teacher ratios is invalid. To give one example only, Blatchford and various colleagues (2003, Blatchford et al., 2009; Blatchford et al., 2011) conducted a series of large-scale naturalistic studies of teaching, learning and achievement in both primary and secondary classrooms in England. Naturalistic studies examine classrooms as they are. As designed by Blatchford and colleagues, the studies focused in fine-grained detail on what occurs in classes of many different sizes using multiple quantitative and qualitative methods. They provided a rich and nuanced picture of complex classroom life and social interaction. They followed both cohorts of students and targeted individuals across several years. They used sophisticated multi-level statistical modelling techniques to identify, examine and determine the weight of relationships between numerous educational variables.

As a coherent programme of empirical classroom research, these studies have considerable policy and practice value across the compulsory schooling sector. It was demonstrated, for example, that larger classes produced more groups of students in the class and larger numbers of students within the groups. This had negative effects on teaching, learning and students' concentration. In smaller classes, teachers were more likely to spend time with individual students. In smaller classes, students were more likely to be engaged in learning and less disruptive; in larger

classes they are more likely to just passively listen to the teacher; in smaller classes students actively interacted with the teacher about their learning. While small classes reportedly have demonstrable benefits for all students in terms of teacher attention, the researchers argue that they may be particularly beneficial in the first years of primary and secondary school and, in the latter, especially for lower-attaining students who otherwise have been shown to be more likely to disengage from learning.

In short, the research showed that class size differences demonstrably affected: (i) the interactional framework of the classroom as a whole (the size and number of within-class groups, curriculum coverage and the quality of students' work); (ii) the teacher (task time with students, individual support for learning, classroom management and control, teacher stress/compensatory efforts); and (iii) the student (active involvement with the teacher, student attentiveness/off-task behaviour, peer relations).

Conclusion

A balanced assessment of different, complementary kinds of research evidence on class size effects reveals that the Treasury's briefing recommendation is at best incomplete, and at worst, potentially harmful. If implemented, the research evidence suggests it is likely that any potential harm would most adversely affect the learning and achievement of the Ministry of Education's priority groups of students, those who are already materially disadvantaged in terms of their education and life chances. This would be unjust.

Despite some overall improvements, the gap between our high performing and low performing students remains one of the widest in the Organisation of Economic Cooperation and Development (OECD). These low performing students are likely to be Maori or Pasifika and/or from low socio-economic communities. Disparities in education appear early and persist throughout learning. (Ministry of Education, 2011, p. 8)

The consistent position of the National-coalition government since 2008 has been that it is possible to address this challenge within schools by raising achievement standards in literacy and numeracy, ensuring that lower-achieving students are supported to remain sufficiently engaged and thereby gain secondary school credentials; and meeting the particular needs of Māori and Pasifika students. Treasury asserts that it is possible to increase student-teacher ratios in schools without compromising this complex challenge. However, the evidence on which this assertion is based is seriously flawed because it fails to take into account what happens to real teaching and learning processes, and student achievement in larger classes.

New Zealand's student-teacher ratios are presently higher than the OECD mean in primary and lower secondary schools, which implies that average class sizes are also larger than the OECD mean. Naturalistic studies of class size effects on teaching and learning clearly demonstrate multiple links between class size and classroom

processes. These same studies suggest that larger class sizes have the most adverse consequences for the achievement of the very groups of learners the Ministry of Education states are its priorities. There is no evidence-based justification to increase student-teacher ratios in schools. On the contrary, OECD data suggests that priority groups of learners could benefit if ratios were to be reduced further in primary and lower secondary schools.

Analysis of the shortcomings of policy advice texts (in this instance, lack of clarity of terminology, incompleteness of evidence and invalid argumentation) only takes us so far in understanding the political economy of state schooling policy. As suggested earlier in this commentary, in order to gain a richer and more nuanced set of understandings of the uses and misuses of education research evidence and policy advice, one needs to focus also on the subject positionings of the various policy actors within contemporary discourses of schooling reform and, as a result, which subjects or positions are heard, and which silenced. Cases such as this suggest that, on the one hand, researchers have a particular responsibility to be watchful both for material fallacies and their mistaken use by policy actors, and on the other hand, they must be prepared to interrupt and challenge flawed public education policy debates whenever they occur.

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